SAM CHRISTOBER RAJA K

Embedded Firmware Engineer | IoT Systems

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 3.5+ years of experience in developing production-grade embedded systems on ESP32, ARM Cortex-M, and Raspberry Pi platforms for industrial automation and connected edge applications. Skilled in Embedded C/C++, FreeRTOS, device driver development, low-power optimization, OTA updates, and industrial communication protocols including SPI, I2C, UART, and RS485/Modbus RTU. Experienced in end-to-end firmware development, hardware bring-up, field deployment, debugging, and scalable embedded product development.

TECHNICAL SKILLS

Languages: Embedded C, C++, C, Arduino (C/C++-based Programming).
Microcontrollers / SoCs: ARM Cortex-M3, ESP32, ESP32-C3, ESP32-CAM, Raspberry Pi, Arduino Boards
RTOS: FreeRTOS (tasks, queues, semaphores, timers, mutexes)
Communication Protocols: UART, SPI, I2C, RS485, Modbus RTU, TCP/IP, MQTT, HTTP/REST
Wireless Technologies: LoRa, Lora WAN, Wi-Fi, ESP-NOW, GSM/GPRS, LTE, NB-IoT
Firmware Concepts: Interrupt handling, DMA, bootloaders, OTA updates, device drivers, low-power optimization, watchdog timers
Debugging & Tools: Serial/JTAG debuggers, Git, VS Code + Platform IO, Arduino IDE, Keil MDK-5.
Cloud & Connectivity: MQTT broker, REST APIs, cloud telemetry pipelines.
Hardware Components: RTC, SD card module, LoRa SX1262, Relay modules, LCD/OLED Display, LEDs, GPS
Sensors: DHT22, BH1750, Soil sensors, Air quality sensors, Proximity Sensors.Etc.,
Concepts: Low-Power Design, Firmware Optimization, Hardware Bring-up, IoT Systems
Database: SQL Server Management.

CORE COMPETENCIES

Embedded Firmware Development | FreeRTOS | Device Drivers | Hardware Bring-Up | Low-Power Optimization | Industrial Protocols | OTA Updates | Firmware Validation | Production Deployment | Linux Device Drivers

PROFESSIONAL EXPERIENCE

Senior IoT / Embedded Firmware Engineer | Sun Bio IT Solutions, Bangalore | *Sep 2022 – Present*

- Engineered and deployed firmware for 25+ IoT products on ESP32, ARM Cortex-M3, Raspberry Pi, and Arduino platforms for industrial automation and smart agriculture applications.
- Developed multi-threaded FreeRTOS-based embedded systems with optimized task scheduling and power management, reducing power consumption by up to 30% in field devices.
- Implemented Lora WAN (Class A/C), Wi-Fi, and GPRS/LTE communication stacks with OTA firmware updates, enabling scalable remote monitoring and maintenance for 50+ deployed units.
- Designed low-level drivers and protocols including SPI, I2C, UART, RS485, and Modbus RTU for seamless integration with industrial sensors and controllers.
- Created reusable firmware libraries, technical documentation, and deployment workflows, accelerating product development and improving maintainability.
- Collaborated directly with clients and field teams for technical presentations, on-site commissioning, troubleshooting, and successful project delivery.

PROJECTS

Industrial Precision Irrigation Control System

Role: Senior Embedded Firmware Engineer

Technologies: *ESP32, LoRa, Lora WAN, FreeRTOS, MQTT, Wi-Fi (HTTP/TCP), LTE, OTA, SPI, UART, I2C*

- Developed end-to-end firmware for multi-node agricultural IoT systems on **ESP32**, integrating soil and environmental sensors (**moisture, temperature, humidity, EC, pH, NPK**) via **UART (RS-485**) and digital interfaces, with **FreeRTOS-based task scheduling** for reliable real-time data acquisition.
- Implemented **Lora WAN and MQTT-based communication**, **automated irrigation control**, and low-power optimization (**deep sleep**), achieving ~3 km range and multi-month battery life while ensuring robust data transmission and zero-loss operation during network interruptions.

Automated Railway Track Lubrication Control System

Role: Senior Embedded Firmware Engineer

Technologies: ESP*32, GSM/GPRS, Modbus RTU, RS485, WIFI (TCP/HTTP), MQTT, OTA, ESP32-CAM, LTE*

- Developed embedded firmware for an **automated grease dispensing system**, enabling precise axle-based lubrication using interrupt-driven inductive sensing with debounce filtering for accurate operation at speeds up to 80 km/h.
- Integrated **Modbus RTU (RS485) and GSM/GPRS** telemetry for robust communication and real-time remote monitoring, ensuring reliable performance and fault reporting in harsh railway environments.

Autonomous Embedded Control System for Solar Maintenance

Role: Senior Embedded Firmware Engineer

Technologies: ESP*32-C3, ESP-NOW, FreeRTOS, PWM Motor Control, MQTT, OTA, LAN, LTE*

- Developed **ESP32-C3** based firmware for a wireless solar panel cleaning robot using **ESP-NOW**, enabling low-latency (<10 ms) peer-to-peer control over ~50 m without infrastructure dependency.
- Implemented **FreeRTOS-based** motor control (PWM for BLDC, servo steering) and custom communication protocol, delivering smooth motion control with configurable speed profiles.

Embedded Electrochemical Water Quality Monitoring System

Role: Senior Embedded Firmware Engineer

Technologies: ARM Cortex -M3 (ADUCM355), *ESP32, FreeRTOS, SPI, UART, WIFI (TCP/HTTP)*

- Developed embedded firmware for a water quality monitoring system using **ADuCM355** (AFE technology) to acquire electrochemical measurements (**DPV and SWV**) for precise analysis of water parameters.
- Implemented **Wi-Fi and Bluetooth connectivity** for real-time data transmission to mobile applications, using **Keil MDK for firmware development and SWD-based debugging and validation**.

EDUCATION

Bachelor of Engineering – Computer Science & Engineering

DMI College of Engineering, Chennai | 2018 – 2022 | CGPA: 8.51 / 10.0